Internship Report

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Abstract

This report describes what I did over the internship at Shanghai Jiao Tong University. I mainly did the following:

I built the benchmark, PointMass3D, with an expert pipeline over RRT-Connect, CHOMP and TrajOpt; then a conditional flow-matching planner over whole trajectories; then the symmetry study.

Varying only the representation raises the collision-free rate from %14.60 to %51.10 at convergence over three seeds, a gap of +36.5 points, and the gap *widens* with training data rather than shrinking: across a 12.5× range of environments at a fixed budget it goes from +6.4 to +30.1. All three mechanisms built for the sixth degree of freedom are worth almost nothing at convergence: roll augmentation $+0.8 \pm 0.3$, frame averaging +0.68, and an exactly SO(2)-equivariant backbone +0.15, seed-matched. The architecture is not pointless, much faster (2x to 3x) than the unconstrained model, so the symmetry buys time, not accuracy. I also bound my own contribution from above: supplying each waypoint with the signed distance to the scene and its gradient is worth +40.0 points where the reduction is worth +36.5, so the representation change studied here is not the largest effect available on this benchmark, and yet, with that information given, the canonicalization is still worth +24.8. I explain this by measuring the *non-equivariance residual*: only about 1.7% of the learned velocity field lies outside the equivariant subspace, and frame averaging is exactly the orthogonal projection onto it, so that 1.7% is the whole budget any post-hoc symmetrisation of this model can spend. Two further findings support the reading: representation and optimisation are separate bottlenecks, and the whole pattern replicates in a second state space where a state is a full rigid-body pose.

1 What I did, in the order in which I did it

The internship ran in three stages, and this report follows them.

benchmark. I built PointMass3D: a cluttered 3D workspace with an analytic signed distance field, and an expert pipeline combining RRT-Connect, random shortcutting, uniform resampling and CHOMP refinement. Implementing three classical planners rather than one also gave me the *ceiling* the learned planner should be read against.

the baseline planner. I also built a conditional flow-matching model over whole trajectories, and the calibration that turned out to matter most: measuring what a straight line and an untrained stochastic prior achieve on the identical problems.

the symmetry study. The $(\mathbf{s}, \mathbf{g})$ reduction, the two obstructions that rule out any continuous gauge for the sixth degree of freedom, the three mechanisms built for it, and after doing everything and finding not much, explaining why they are worth little at the budget of the main grid, and why that qualifier matters.

1.1 Contributions

(i) Five of six degrees of freedom are free. The query supplies a frame. Placing the origin at the midpoint of $\mathbf{s}$ and $\mathbf{g}$ and aligning the first axis with the start–goal direction removes three translational and two rotational degrees of freedom exactly, at no computational cost and with no architectural constraint. The resulting representation is invariant under any rigid motion of the problem *up to a single roll about the start–goal axis.*

(ii) No continuous rule fixes the sixth, and the natural scene-derived alternative carries no signal. A continuous rule fixing the roll would be a nowhere-vanishing tangent field on S^2, which the hairy ball theorem forbids. Deriving the roll from the scene instead fails because the obstacle law is symmetric under central inversion, so the angular moment such a gauge would use vanishes in expectation.

(iii) The symmetry machinery buys little, and I measured why. Rather than infer inertness from flat quality curves, I measured the quantity frame averaging exists to remove, and proved exactly what it bounds.

(iv) Calibration, and two bottlenecks. A collision-free rate on a new benchmark means nothing without a scale. Measuring the floor is what revealed that the world-frame arm sits exactly on the straight line, and measuring a longer training run is what revealed that compute and representation relieve different constraints.

I regard (iii) as the transferable part. The residual is a few lines of code, costs one forward pass, needs no retraining, and can be computed on any model with a group action on its inputs. It converts "we tried equivariance and it did not help" into a quantitative statement with an exact bound attached — and it comes with an equally exact statement of its limits, which Sec. 6.4 makes concrete.

2 Stage 1: The PointMass3D benchmark

The workspace is $[-1, 1]^3$ holding forty obstacles, twenty spheres and twenty oriented boxes, with an analytic signed distance field so collision checking is exact rather than sampled. The robot is a point of radius 0.03. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$.

Expert paths come from RRT-Connect, then random shortcutting, uniform resampling, and CHOMP refinement. I implemented TrajOpt as well. It was not needed for the dataset, but it gave me two more points on the ceiling the learned planner should be read against, and that turned out to matter more than the dataset did.

The released set is 300 environments × 36,000 trajectories, from 600 start–goal pairs each with 30 paths and their time reversals. Environments 250–299 are held out and never trained on. Thirty near-duplicate paths per pair means the nominal ten million trajectories carry on the order of 360,000 distinct queries. This affects no comparison here, since both arms see identical data, but if I generated more I would cut paths per pair and buy environments instead.

3 Stage 2: The flow-matching planner

The planner is a conditional flow-matching model over whole trajectories. Training draws $x_0 \sim \mathcal{N}(0, I)$ and $t \sim U[0, 1]$, forms $x_t = (1 - t)x_0 + t\, x_1$ with x_1 an expert trajectory, and regresses the velocity $x_1 - x_0$. Sampling integrates the learned field from noise with eight Euler steps.

The backbone is a dilated temporal convolutional trunk with FiLM conditioning and a PointNet-style obstacle encoder, 2.16M parameters. The encoder max-pools all forty obstacles into a single 128-dimensional vector, and FiLM applies that vector identically to every one of the sixty-four waypoints.

I considered a Brownian-bridge prior, which would start sampling from paths that already join the endpoints, and did not adopt it. The reason I first gave was wrong. I thought the bridge was not isotropic in the plane normal to the start–goal axis; it is. The decision survives for a different reason: the bridge is query-dependent and its endpoint covariance is singular, which complicates the comparison this report exists to make.

4 Stage 3: Method

4.1 The query already supplies a frame

A planning problem is unchanged if you pick it up and move it. For any rigid $T \in \mathrm{SE}(3)$, the solution to the transformed problem is the transformed solution. A network trained on world coordinates cannot see this, and must learn the same manoeuvre again at every position and orientation.

The start and goal define a frame with no learning at all. Place the origin at the midpoint of $\mathbf{s}$ and $\mathbf{g}$, align the first axis with the start–goal direction, and express every obstacle and every waypoint in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at no computational cost and with no architectural constraint. The six numbers used to condition the network on the query collapse to one invariant scalar, the separation d.

The frame needs a second axis to be fully determined, and mine takes it from a fixed world axis.

What survives, precisely. I had this wrong for several weeks. The reduced representation is *not* invariant under SE(3): rotating the whole problem selects a different tie-break axis and rolls the frame. What is true is that the entire residual is that one roll. Every reduced quantity, waypoints, sphere centres and box edge vectors alike, maps through one shared rotation about the first axis, verified to 10^{-15}. The reduction collapses SE(3) to SO(2), not to nothing. Invariant outright are d, the first coordinate of every reduced point, and its radius in the plane normal to the axis.

If the representation had been fully invariant there would be no roll left to handle, and the rest of this section would be unmotivated.

4.2 No continuous rule fixes the sixth degree of freedom

Obstruction from geometry. A rule assigning a perpendicular direction $\hat{\mathbf{y}}(\hat{\mathbf{x}})$ continuously over all axis directions is a nowhere-vanishing tangent field on S^2, which the hairy ball theorem forbids. The start–goal directions in this dataset cover the whole sphere, so no contractible-domain escape is available. Every deterministic gauge therefore jumps somewhere. For the smallest-component cross-product construction I use, the jump is 60 to 120 degrees.

Obstruction from the scene. One might derive the roll from the obstacles instead, pointing the second axis at the obstacle mass. That quantity is the first angular moment of the obstacle distribution. The obstacle law is symmetric under $\mathbf{p} \mapsto -\mathbf{p}$, which preserves every radius and sends $\varphi \mapsto \varphi + \pi$, so every odd moment vanishes in expectation for every axis direction. Even moments survive this argument. They do not survive the first one: a second-moment gauge defines a director field, and a continuous director field on S^2 is obstructed for the same Euler-characteristic reason.

Both propositions forbid *continuous* gauges. A discontinuous rule exists and I use one. The sixth degree of freedom therefore has to be handled by symmetry rather than by canonicalisation.

4.3 Three ways to handle the leftover roll

A symmetry can be imposed in exactly three places, and I built all three.

In the data. Roll augmentation: apply a uniform random roll to each training example, so the network sees every orientation of the residual.

In the operator. Frame averaging: at inference, evaluate the velocity field at K equally spaced rolls, map each result back, and average. The sampled field is then exactly C_K-equivariant whatever the network learned.

In the weights. An SO(2)-equivariant architecture, in which equivariance holds by construction rather than by training (Sec. 6.3).

4.4 The non-equivariance residual, and what it bounds

Rather than infer that these mechanisms are inert from flat quality curves, I measured the quantity frame averaging exists to remove. Let f be the learned velocity field and $\mathcal{A}f$ its average over K rolls. The non-equivariance residual r is the RMS disagreement among the rotated copies, normalised by the field magnitude. It costs one forward pass and needs no retraining.

$\mathcal{A}$ is linear, idempotent and self-adjoint under the natural inner product, so it is the orthogonal projection onto the C_K-equivariant subspace. The target field is itself equivariant, so Pythagoras bounds the error frame averaging can remove by r^2. Measured across four data scales, r is 1.7% and flat. The entire budget available to any post-hoc symmetrisation of this model is therefore under 3×10^{-4}, while the model is failing on most of its samples. The model's error is overwhelmingly equivariant error.

What r bounds is what symmetrising a *given* field can remove. Sec. 6.4 shows what it does not bound.

4.5 Implementation, and the errors worth recording

Oriented boxes. The reduction is a general rotation, so axis-aligned boxes do not stay axis-aligned. I changed the box representation to a centre plus three half-edge vectors, and gave both arms that representation so it is not a confound.

Points and free vectors transform differently. The reduction is affine. Waypoints, start, goal and obstacle centres take the full affine map. The three half-edge vectors take the rotation only. Applying the affine map to an edge vector leaves box positions correct and destroys box extent, and the loss curve looks healthy throughout. The same class of error appeared later in the rigid-body domain's rotation columns.

Verification. Sixty-one tests across six files, all property-based. Every equivariance test runs on an untrained network, which is the case that shows the property comes from the structure of the weights rather than from training. Each also carries a negative control, since a network that ignored its vector inputs entirely would be trivially equivariant. The tests check that the model is *not* equivariant to rotations off the relevant axis.

5 Protocol and calibration

Every arm is evaluated on the same 500 held-out problems: 50 environments $\times$ 10 distinct start–goal pairs $\times$ 20 samples, eight Euler steps. I report two metrics and never conflate them. A sample is *free* if the whole path is collision-free. *Best-of-20* is whether any of the twenty samples for a query is free. They differ by about thirty points.

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	<0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	14.6	−0.071	0.053
flow, $(\mathbf{s}, \mathbf{g})$ reduction	**51.1**	−0.002	0.063

Table 1: The same 500 held-out problems solved five other ways. The learned rows are the converged 60-environment arms, three-seed means. Classical timings are one CPU core and one query; the learned rows are one GPU and twenty samples, so the time column is not like-for-like.

Training environments	20	60	150	250
world frame	12.86	13.57	14.75	15.37
$(\mathbf{s}, \mathbf{g})$ reduction	19.23	34.43	40.60	**45.45**
gap	+6.4	+20.9	+25.9	+**30.1**

Table 2: A fixed 20-epoch budget at four data scales, three seeds per cell.

Uncertainty is quoted across seeds for claims about a method, since that is what varies when the method is rerun. For comparisons against a fixed baseline measured on the same problems I use a paired bootstrap over environments instead.

The floor. A straight line from start to goal is collision-free on 15.6% of these problems. The world-frame model, trained to a plateau, converges to $14.60 \pm 0.20\%$.

I first wrote that it was significantly below the line, quoting the across-seed spread. That was the wrong denominator: the spread describes variation between training runs, not the uncertainty of a comparison against a fixed baseline on the same problems. The right test is paired. Resampling environments and differencing within each replicate gives −1.02 points, a 95% interval of $[-3.06, +0.90]$, and a standard error of 1.01, less than half what either rate carries alone. What I can defend is that world coordinates do not beat a straight line, not that they lose to one.

Which number is the headline. Three levels appear in this report. 51.10 ± 0.25 against 14.60 ± 0.20 is the headline: sixty environments, trained to a plateau, three seeds, and the setting every comparison below uses. The 45.5 in the scaling grid is a fixed 20-epoch budget, which exists to compare data scales at equal optimisation. The 55.8 at 250 environments is the largest number I measured and rests on a single seed.

6 Results

6.1 The reduction scales; the baseline does not

The objection I expected most was that canonicalisation is a small-data crutch. The gap widens monotonically with data. The control gains 2.6 points across a 12.5× increase in layouts; the reduction gains 26.2 and has not plateaued at the dataset cap.

Trained to convergence at sixty environments the gap is $+36.5 \pm 0.32$ points. The control gains nothing from a 15× budget increase: it moves 14.6 to 14.4 and stays where it started. Minimum clearance agrees independently, the control sitting at −0.07 throughout while the reduction approaches feasibility at −0.002.

Mechanism	Worth	
$(\mathbf{s}, \mathbf{g})$ reduction (5 DOF)	+36.5	± 0.32, $n = 3$
of which: 3 translations	+12.3	recentring
of which: 2 rotations	+18.0	axis alignment
roll augmentation (data)	+0.8	± 0.3 paired, n.s.
frame averaging (operator)	+0.68	± 0.13, $n = 3$
SO(2)-equivariant net (weights)	+0.15	seed-matched, $n = 3$

Table 3: The five removable degrees of freedom against the three mechanisms built for the sixth.

Epoch	equivariant	unconstrained	Δ
10	35.39	22.25	+13.14
40	44.91	38.17	+6.74
100	48.82	45.98	+2.84
300	51.27	51.12	+0.15

Table 4: Sixty environments, identical data and schedule, seed-matched.

6.2 All three mechanisms for the sixth degree of freedom

Three unrelated ways of imposing the same symmetry all recover under a point. That is a stronger statement than any one of them makes alone. For most of the internship I had only the first two, and was arguing from the flatness of their curves.

6.3 The third mechanism: I built the equivariant architecture

I had argued from the smallness of r that a constrained architecture would gain little. Sec. 6.4 shows that inference is unsound, so I built it.

The backbone is exactly SO(2)-equivariant about the start–goal axis by construction. Features are separated by irreducible representation: the component along the axis is invariant, and the pair in the normal plane rotates together as one complex number. Only operations respecting that separation are allowed, namely complex-linear channel mixing, obstacle pooling with attention weights computed from invariants alone, normalisation of the rotating stream by an invariant, and gating of that stream by scalars. Equivariance is verified on an untrained network at 1.2×10^{-6}, and survives multiplying every parameter by three. I did not match capacity downward. The constrained model carries 2,582,233 parameters against 2,161,283, or 19.5% more, because a constrained model that is also smaller confounds architecture with capacity.

The constraint buys optimisation and not accuracy. The equivariant model reaches by epoch 10 a level the unconstrained arm needs about 32 epochs for, and by epoch 40 one it needs about 87. At convergence the difference is +0.15 points, which is zero.

A bug worth recording. My first implementation plateaued at 22.2%, and the mathematics was correct throughout. Every equivariance test passed at 10^{-6}. The fault was in how information moved through it. I pooled the obstacles' rotating features by taking their mean, and the mean of a roughly symmetric cloud of directions is close to nothing. I measured that it retained 14% of a typical obstacle's magnitude. The one channel carrying which way the obstacles lie was arriving empty. Correct and useless are different failure modes, and only one of them has a test.

Arm (250 environments)	r under SE(3)	Free (%)
world frame	0.0881	15.4
+ SE(3) augmentation	0.0181	17.5
$(\mathbf{s}, \mathbf{g})$ reduction	0.0174	**45.6**

Table 5: Augmentation and canonicalisation leave the field equally equivariant.

60 envs, converged	global only	+ local	geometry adds
world frame	14.60	54.54	+40.0
$(\mathbf{s}, \mathbf{g})$ reduction	51.10	79.37	+28.3
+ equivariant backbone	51.27	**82.13**	+30.9
reduction contributes	+36.5	+24.8	

Table 6: Every cell is a mean over two or three seeds. The local-geometry arms are the least seed-sensitive measurements in this work.

6.4 What the residual does not predict

Augmentation drives the residual down by a factor of 4.9. It genuinely teaches the network the symmetry, to the same degree the reduction enforces it by construction, and it buys 2.1 points. The reduction, at an indistinguishable residual, buys 30.1. So r does not forecast what a mechanism is worth, and the rule "build the constrained architecture only if r is large" has no support.

The world-frame arm performs at the straight-line floor. It is 91% equivariant.

6.5 Is the benefit simply letting the encoder see the query?

Linear probes give the world-frame scene code a within-environment standard deviation of 0.000000. The encoder sees only obstacles, so its output cannot vary with the query, and the reduced code by contrast varies almost entirely within an environment. That invites a deflationary reading of the whole report: perhaps the reduction buys query-dependence rather than canonicalisation, and a one-line change to the encoder would buy the same thing.

I built the arm that separates them. It is the world frame throughout, with no change of frame anywhere, but the raw query is concatenated to every obstacle before the pool, so its scene code does vary with the query. It carries 0.07% more parameters. It converges to $15.59 \pm 0.38\%$ against the world frame's 14.60 ± 0.20, both over three seeds, and seed-matched the difference is +1.01 and positive on every seed. Query-dependence alone is worth about one point of the 36.5. Canonicalisation supplies the other 35.5.

At 15.59% the arm sits on the straight line's 15.6%. Giving the encoder the query is exactly enough to reach the trivial baseline and not enough to beat it.

6.6 Local geometry, and how much of my result it explains

The obstacle encoder pools forty obstacles into one vector applied identically to all sixty-four waypoints, so an individual waypoint has no channel through which to ask what is near it. Collision avoidance is that question. I appended to each waypoint the signed distance to the scene and its gradient. This is a featurisation rather than an oracle, since the signed distance is a deterministic function of obstacle primitives the network already receives. What it assumes is that obstacles are available as explicit primitives at inference.

Local geometry is worth +40.0 points against the world frame. The reduction is worth +36.5. The representation change this report is about is not the largest effect available on its own benchmark.

If the reduction were only a workaround for a weak encoder, supplying per-waypoint geometry should absorb its benefit. It does not. With local geometry present the reduction is still worth +24.8 points. I ran that cell specifically to find out whether it would overturn my main result.

The margin does decay with data. Comparing at a matched budget and differencing within seed, the reduction alongside local geometry is worth $+22.37 \pm 0.46$ at sixty environments and $+17.24 \pm 0.17$ at 250. Anyone extrapolating to a much larger dataset should expect it to narrow rather than hold.

The equivariant backbone is worth +0.15 points without local geometry and +2.8 with it. The rotating stream has nothing to act on until directional information reaches individual waypoints.

6.7 Three shorter results

Two bottlenecks, not one. Tripling the training budget is worth +8.0 points to the canonicalised arm and +0.9 to the world-frame one, an interaction of +7.1. Compute cannot buy what the representation forecloses. On the best-of-20 metric the same tripling moves the world-frame arm backwards, from 42.3% to 39.2%. More training bought a per-sample gain while costing sample diversity.

It replicates in a second state space. I built an SE(3) rigid-body domain whose state is a full pose rather than a point, using an L-shaped union of five spheres chosen to have trivial symmetry so the domain is not secretly easier. The world-frame arm fails to clear its own trivial floor in both domains, the reduction clears it in both, and the ratio between arms is $2.9\times$ in each.

It is not specific to flow matching. Replacing the objective with DDPM and holding everything else fixed reproduces the gap: +37.0 at convergence against flow matching's +36.5. The two objectives disagree early and agree at convergence.

7 Corrections

Five claims I made during the internship turned out to be wrong. In each case the corrected statement is more useful than the original.

1. I thought the reduced representation was bit-identical under rigid motion. It is not. It is invariant up to one roll, and that correction is what motivates half of this report.

2. I reported roll augmentation as worth +4.4 points. An artefact of comparing a three-seed mean against a single-seed ablation. Seed-matched it is $+0.8 \pm 0.3$ and not significant. The across-seed spread is five times the paired spread, so any ablation compared unpaired at this scale is measuring seed noise.

3. I argued the roll gauge could be fixed by a C_4 argument. That holds only when the axis is a four-fold axis of the cube. The correct hypothesis is central inversion, which is weaker, true, and holds for a much broader class of obstacle distributions than my own generator.

4. I reported the residual as falling with scale. That came from two points under an older definition. Measured at four scales under the definition that makes the bound an identity, it is flat. I had built an argument on the trend, and the argument does not survive.

5. I decided not to build the equivariant architecture, and gave two reasons, both wrong. I argued that r was small and falling, so the architecture would enforce something arriving on its own. The trend was an artefact and the level does not predict. When I finally built it the ceiling was unchanged. I had reached the right answer by an argument that does not support it, and missed the two-to-threefold saving in optimisation entirely.

8 Discussion and limitations

Which robots this transfers to. The reduction needs the group to act on the state space, not just the workspace. That holds for a point mass and for the rigid body of the second domain. It fails for a manipulator planning in joint space: the query still defines a frame in the workspace, but a rigid motion of the world induces no clean action on joint angles. Mobile bases, free-flying bodies and end-effector-space planning pass this test. Joint-space planning for arms does not.

The learned planner loses to the classical ones. RRT-Connect solves 99.6% in under 0.3 s on one CPU core, and the learned sampler reaches 51.1%. On this benchmark one should use RRT-Connect. The contribution is a controlled measurement of what a representation change is worth, and that measurement is only isolable where the confounds are removable.

Scope. One workspace family, one clutter generator, one architecture. The claim that the sixth degree of freedom needs no treatment relies on the training distribution supplying implicit roll diversity through many start–goal directions. A dataset with few directions per environment could plausibly reverse it.

Seeds. The 60-environment cells carry three seeds, the local-geometry arms two or three, and the 250-environment cells one. Every claim I lean on has at least two.

9 What I would do next

Make r vary deliberately. Training on a narrow cone of start–goal directions should starve the model of implicit roll diversity and drive r up. If frame averaging then pays in proportion, r becomes a decision rule for post-hoc symmetrisation. If it does not pay even at large r, that is the harder negative. I attempted this and the attempt failed: narrowing the cone raises r and degrades quality at the same time, and I could not separate them.

Capacity as a controlled variable. The model is small, and "the model is small" is currently a limitation rather than something I measured. Doubling the trunk on both arms would turn it into a controlled variable.

More environments. The scaling curve has not plateaued at 250, which is the dataset cap. Cutting paths per pair from thirty to five would buy roughly six times the environments for the same generation cost.

10 Conclusion

Of the six degrees of freedom of SE(3), five are removable exactly and for free by a frame the query already supplies. That is worth +36.5 points on this benchmark, against a world-frame arm that does not beat a straight line. The sixth is not removable by any continuous rule, and all three mechanisms available for it are worth under a point each. The architecture is not pointless: it reaches any given level two to three times sooner, so the symmetry buys optimisation rather than accuracy.

Supplying per-waypoint geometry is worth more than the frame is, and the frame still adds +24.8 points alongside it. The recommendation I would make for this benchmark is correspondingly narrow. Canonicalise the degrees of freedom that admit a continuous canonical form, because the query usually supplies more of them than one expects, and measure the rest rather than assuming them.